

KUNCI_LATIHAN_Pertemuan11

KUNCI JAWABAN - SOAL LATIHAN PERTEMUAN 11

Greedy, D&C, Backtracking - Fase F

A. GREEDY

Soal 1 — Penukaran Koin Rp23.400

Langkah	Sisa Uang	Ambil Pecahan	Sisa Baru
1	Rp23.400	Rp10.000 (2×)	Rp3.400
2	Rp3.400	Rp2.000	Rp1.400
3	Rp1.400	Rp1.000	Rp400
4	Rp400	Rp200 (2×)	Rp0

Jumlah minimal: 6 lembar/koin ($2 \times 10.000 + 2.000 + 1.000 + 2 \times 200$)

Soal 2 — Fractional Knapsack 40 kg

Density: - A: $120\text{jt}/8 = \text{Rp}15\text{jt/kg}$ - B: $150\text{jt}/15 = \text{Rp}10\text{jt/kg}$ - C: $180\text{jt}/12 = \text{Rp}15\text{jt/kg}$ - D: $80\text{jt}/10 = \text{Rp}8\text{jt/kg}$ - E: $90\text{jt}/6 = \text{Rp}15\text{jt/kg}$

Urutan density: C (15jt), A (15jt), E (15jt), B (10jt), D (8jt)

Urutan	Barang	Density	Berat	Ambil?	Total Berat
1	C	15jt	12	12 kg (ambil semua)	12
2	A	15jt	8	8 kg (ambil semua)	20
3	E	15jt	6	6 kg (ambil semua)	26
4	B	10jt	15	14 kg (sisa kapasitas)	40
5	D	8jt	10	0 (penuh)	40

Total nilai: C: $12 \times 15\text{jt} = \text{Rp}180\text{jt}$ A: $8 \times 15\text{jt} = \text{Rp}120\text{jt}$ E: $6 \times 15\text{jt} = \text{Rp}90\text{jt}$ B: $14 \times 10\text{jt} = \text{Rp}140\text{jt}$ **Total = Rp530jt**

B. DIVIDE & CONQUER

Soal 3 — Binary Search

Cari 35: | Langkah | Kiri | Kanan | Tengah | Nilai | Aksi | |—|—|—|—|—|—| | 1 | 0 | 9 | 4 | 24 | $35 > 24 \rightarrow$ kiri = 5 | | 2 | 5 | 9 | 7 | 42 | $35 < 42 \rightarrow$ kanan = 6 | | 3 | 5 | 6 | 5 | 29 | $35 > 29 \rightarrow$ kiri = 6 | | 4 | 6 | 6 | 6 | **35** |
Ditemukan! |

Indeks: 6

Cari 9: | Langkah | Kiri | Kanan | Tengah | Nilai | Aksi | |—|—|—|—|—|—| | 1 | 0 | 9 | 4 | 24 | 9 < 24 → kanan = 3 | | 2 | 0 | 3 | 1 | **9 | Ditemukan!** |

Indeks: **1**

Soal 4 — Merge Sort

Divide:

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[54, 23, 11, 38, 47, 6, 29, 15]
[54, 23, 11, 38]    [47, 6, 29, 15]
[54, 23]    [11, 38]    [47, 6]    [29, 15]
[54] [23] [11] [38] [47] [6] [29] [15]
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Combine:

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[23, 54]    [11, 38]    [6, 47]    [15, 29]
[11, 23, 38, 54]    [6, 15, 29, 47]
[6, 11, 15, 23, 29, 38, 47, 54]
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Hasil akhir: [6, 11, 15, 23, 29, 38, 47, 54]

C. BACKTRACKING

Soal 5 — Permutasi {A, B, C, D} — A tidak boleh sebelum B

Pohon Backtracking dengan pruning:



Semua permutasi dari {A,B,C,D} = 24. Syarat A tidak sebelum B.

Permutasi valid (A muncul setelah B): - B harus sebelum A dalam urutan - Dari 24 permutasi, setengahnya (12) memiliki A sebelum B, setengahnya (12) memiliki B sebelum A

Permutasi valid (12): B, A, C, D — B, A, D, C — B, C, A, D — B, C, D, A — B, D, A, C — B, D, C, A — C, B, A, D — C, B, D, A — C, D, B, A — D, B, A, C — D, B, C, A — D, C, B, A

Soal 6 — Labirin 4×4

	Kolom 0	1	2	3
Baris 0	S	.	[X]	.
Baris 1	[X]	.	.	[X]
Baris 2	.	[X]	.	.
Baris 3	.	.	[X]	F

Jalur: (0,0)→(0,1)→(1,1)→(2,1) → backtrack karena (2,1) punya (2,0) dan (1,1) sudah visited, (3,1) bisa tp coba dulu (2,2)...

Wait, let me trace: (0,0) S ↓ (0,1) ↓ (1,1) ↓ (2,1) — [X] × → backtrack
to (1,1) (1,1) → (1,2) ↓ (2,2) ↓ (3,2) — [X] × → backtrack to (2,2) (2,2) →
(2,3) ↓ (3,3) F

Jalur: (0,0)→(0,1)→(1,1)→(1,2)→(2,2)→(2,3)→(3,3) F

Atau alternatif: (0,0)→(0,1)→(1,1)→(1,2)→(2,2)→(2,3)→(3,3)

Let me also check (2,0): (0,0)→(0,1)→(1,1)→(1,2)→... → what about going (0,0)→(0,1)→(1,1)→ no wait, (1,1) can't go to (2,1) since it's [X]. So from (1,1): (0,1) visited, (2,1)=[X] ×, (1,2) → that's the only option.

Actually let me re-check (2,1) — looking at the grid: Baris 2, Kolom 1 is [X].

From (0,0): right → (0,1), down → (1,0)=[X] × From (0,1): right → (0,2)=[X] ×, down → (1,1)
From (1,1): down → (2,1)=[X] ×, right → (1,2) , left → (1,0)=[X] ×, up → (0,1) visited From
(1,2): down → (2,2) , right → (1,3)=[X] ×, left → (1,1) visited From (2,2): down → (3,2)=[X] ×,
right → (2,3) , left → (2,1)=[X] ×, up → (1,2) visited From (2,3): down → (3,3)=F

Jalur: (0,0)→(0,1)→(1,1)→(1,2)→(2,2)→(2,3)→(3,3)F

That's correct!

Rubrik Penilaian

Soal	1 (25%)	2 (50%)	3 (75%)	4 (100%)
1 (koin)	Tidak diisi	Langkah sebagian	6 lembar	6 + langkah lengkap
2 (knapsack)	Tidak diisi	Density benar	Urutan + ambil benar	Total Rp530jt
3 (binary)	Tidak diisi	1 angka benar	2 angka benar	2 + langkah
4 (merge sort)	Tidak diisi	Divide saja	Sebagian combine	[6,11,15,23,29,38,47,54]
5 (permutasi)	Tidak diisi	Pruning sebagian	6 permutasi	12 permutasi + pohon
6 (labirin)	Tidak diisi	Jalur sebagian	Jalur benar	Jalur + backtrack

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